Homework #0

CSE 546: Machine Learning
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Probability and Statistics

A1. *[2 points]* (From Murphy Exercise 2.4.) After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease, and that the test is 99% accurate (i.e., the probability of testing positive given that you have the disease is 0.99, as is the probability of testing negative given that you don't have the disease). The good news is that this is a rare disease, striking only one in 10,000 people. What are the chances that you actually have the disease?

If you test positive for the disease, the probability that you actually have it is approximately 0.98%.

Let + denote the event of receiving a positive test result and - denote the event of receiving a negative test result. Also, let D denote the event of having the disease and $\overline{D}$ the event of not having the disease.
Thus, we have that

$$\mathbb{P}(D) = 0.0001 \Rightarrow \mathbb{P}(\overline{D}) = 1 - \mathbb{P}(D) = 0.9999$$
$$\mathbb{P}(+|D) = 0.99$$
$$\mathbb{P}(-|\overline{D}) = 0.99$$
$$\mathbb{P}(+|\overline{D}) = 1 - \mathbb{P}(-|\overline{D}) = 0.01$$
$$\mathbb{P}(-|D) = 1 - \mathbb{P}(+|D) = 0.01$$

Using Bayes' theorem,

$$\mathbb{P}(D|+) = \frac{\mathbb{P}(+|D)\mathbb{P}(D)}{\mathbb{P}(+)} = \frac{\mathbb{P}(+|D)\mathbb{P}(D)}{\mathbb{P}(+|D)\mathbb{P}(D) + \mathbb{P}(+|\overline{D})\mathbb{P}(D)} = \frac{0.99 \cdot 0.0001}{0.99 \cdot 0.0001 + 0.01 \cdot 0.9999} = \frac{1}{102} \approx 0.0098$$

A2. For any two random variables X, Y the *covariance* is defined as $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$. You may assume X and Y take on a discrete values if you find that is easier to work with.

a. *[1 point]* If $\mathbb{E}[Y \mid X = x] = x$ show that $\text{Cov}(X, Y) = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]$.

Using linearity of expectation,

$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X\,\mathbb{E}[Y]] - \mathbb{E}[Y\,\mathbb{E}[X]] + \mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$

Using the law of total expectation,

$$\mathbb{E}[XY] = \sum_x \mathbb{P}(X = x)\,\mathbb{E}[XY \mid X = x] = \sum_x \mathbb{P}(X = x)\,\mathbb{E}[Y \mid X = x](x) = \sum_x \mathbb{P}(X = x)(x^2) = \mathbb{E}\big[X^2\big]$$

$$\mathbb{E}[Y] = \sum_x \mathbb{P}(X = x)\,\mathbb{E}[Y \mid X = x] = \sum_x \mathbb{P}(X = x)(x) = \mathbb{E}[X]$$

Substituting for $\mathbb{E}[XY]$ and $\mathbb{E}[X]\,\mathbb{E}[Y]$, we have

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}\big[X^2\big] - (\mathbb{E}[X])^2 = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big]$$

b. *[1 point]* If X, Y are independent show that $\text{Cov}(X, Y) = 0$.

We have shown above that $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$.
If X, Y are independent, we have that

$$\mathbb{E}[XY] = \sum_{x,y} \mathbb{E}[XY \mid X = x, Y = y]\mathbb{P}(X = x, Y = y) = \sum_{x,y} \mathbb{P}(X = x, Y = y)(xy)$$

$$\mathbb{E}[XY] = \sum_{x} \mathbb{P}(X = x)(x) \sum_{y} \mathbb{P}(Y = y)(y) = \mathbb{E}[X]\,\mathbb{E}[Y]$$

Therefore, $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[X]\,\mathbb{E}[Y] - \mathbb{E}[X]\,\mathbb{E}[Y] = 0$

A3. Let X and Y be independent random variables with PDFs given by f and g, respectively. Let h be the PDF of the random variable $Z = X + Y$.

a. *[1 point]* Show that $h(z) = \int_{-\infty}^{\infty} f(x)g(z - x)\,\mathrm{d}x$. (If you are more comfortable with discrete probabilities, you can instead derive an analogous expression for the discrete case, and then you should give a one sentence explanation as to why your expression is analogous to the continuous case.).

The joint PDF of X, Y is $f_{X,Y} = f(x)g(y)$.

$$F_Z(z) = \mathbb{P}(Z \leq z) = \mathbb{P}(X + Y \leq z)$$

$$F_Z(z) = \int\int_{x+y\leq z} f_{X,Y}(x, y)dxdy = \int\int_{x+y\leq z} f(x)g(y)dxdy = \int_{-\infty}^{\infty} f(x)\left(\int_{-\infty}^{z-x} g(y)dy\right)dx$$

$$F_Z(z) = \int_{-\infty}^{\infty} F_Y(z - x)f(x)dx$$

$$h(z) = \frac{d}{dz}F_Z(z) = \frac{d}{dz}\int_{-\infty}^{\infty} F_Y(z - x)f(x)dx = \int_{-\infty}^{\infty} f(x)g(z - x)dx$$

b. *[1 point]* If X and Y are both independent and uniformly distributed on $[0, 1]$ (i.e. $f(x) = g(x) = 1$ for $x \in [0, 1]$ and 0 otherwise) what is h, the PDF of $Z = X + Y$?

Considering $h(z)$ on the interval $[0, 1]$, we have

$$h(z) = \int_{-\infty}^{\infty} f(x)g(z - x)dx = \int_{0}^{1} g(z - x)dx$$

$$0 \leq z - x \leq 1 \Rightarrow z - 1 \leq x \leq z$$

For $z \in [0, 1]$:

$$h(z) = \int_{0}^{z} dx = z$$

For $z \in (1, 2]$:

$$h(z) = \int_{z-1}^{1} dx = 2 - z$$

Otherwise, $h(z) = 0$. This gives

$$h(z) = \begin{cases} z & z \in [0, 1] \\ 2 - z & z \in (1, 2] \\ 0 & \text{otherwise} \end{cases}$$

A4. Let $X_1, X_2, ..., X_n \sim \mathcal{N}(\mu, \sigma^2)$ be i.i.d random variables. Compute the following:

a. *[1 point]* $a \in \mathbb{R}, b \in \mathbb{R}$ such that $aX_1 + b \sim \mathcal{N}(0, 1)$.

$$a = \frac{1}{\sigma}, b = \frac{-\mu}{\sigma}$$

We have a random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ which is Gaussian distributed with mean μ and variance σ^2. Let $Y = aX + b$ be another random variable that is also Gaussian distributed where $Y \sim \mathcal{N}(0, 1)$.

$$\mathbb{E}[Y] = a\,\mathbb{E}[X] + b = 0$$

$$a\mu + b = 0$$

$$\mathrm{Var}[Y] = a^2\,\mathrm{Var}[X] = 1$$

$$a^2\sigma^2 = 1$$

Solving this system yields

$$a = \frac{1}{\sigma}, b = \frac{-\mu}{\sigma}$$

b. *[1 point]* $\mathbb{E}[X_1 + 2X_2], \mathrm{Var}[X_1 + 2X_2]$.

Since $\mathbb{E}[X] = \mu$,

$$\mathbb{E}[X_1 + 2X_2] = \mu + 2\mu = 3\mu$$

and

$$\mathrm{Var}[X_1 + 2X_2] = \mathrm{Var}[X_1] + \mathrm{Var}[2X_2] = \sigma^2 + 4\sigma^2 = 5\sigma^2$$

c. *[2 points]* Setting $\widehat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$, the mean and variance of $\sqrt{n}(\widehat{\mu}_n - \mu)$.

$$\mathbb{E}\big[\sqrt{n}(\widehat{\mu}_n - \mu)\big] = \sqrt{n}(\mathbb{E}[\widehat{\mu}_n] - \mathbb{E}[\mu]) = \sqrt{n}\,\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] - \sqrt{n}\,\mathbb{E}[\mu] = \sqrt{n}(\mathbb{E}\left[\frac{1}{n}(n\mu)\right] - \mathbb{E}[\mu]) = 0$$

$$\mathrm{Var}\big[\sqrt{n}(\widehat{\mu}_n - \mu)\big] = n\,\mathrm{Var}[\widehat{\mu}_n] + n\,\mathrm{Var}[\mu] = n\,\mathrm{Var}[\widehat{\mu}_n] + 0 = n\,\mathrm{Var}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = n\frac{\sigma^2}{n} = \sigma^2$$

Linear Algebra and Vector Calculus

A5. Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}$. For each matrix A and B:

a. *[2 points]* What is its rank?

Using Gaussian elimination to obtain the reduced row echelon form of matrix A

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Since there are 2 linearly independent columns, $rank(A) = 2$.

Using Gaussian elimination to obtain the reduced row echelon form of matrix B

$$B = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Since there are 2 linearly independent columns, $rank(B) = 2$.

b. *[2 points]* What is a (minimal size) basis for its column span?

A basis of the column span is composed of the columns that do not pivot when finding the reduced row echelon form.

$$\text{Basis for matrix A's column span} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\text{Basis for matrix B's column span} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

A6. Let $A = \begin{bmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{bmatrix}$, $b = \begin{bmatrix} -2 & -2 & -4 \end{bmatrix}^\top$, and $c = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^\top$.

a. *[1 point]* What is Ac?

$$Ac = \begin{bmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 7 \end{bmatrix}$$

b. *[2 points]* What is the solution to the linear system $Ax = b$?

$$\left[\begin{array}{ccc|c} 0 & 2 & 4 & -2 \\ 2 & 4 & 2 & -2 \\ 3 & 3 & 1 & -4 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & -1 \\ 0 & 2 & 4 & -2 \\ 3 & 3 & 1 & -4 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & -1 \\ 0 & 1 & 2 & -1 \\ 0 & -3 & -2 & -1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 4 & -4 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

$$x = \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}$$

A7. For possibly non-symmetric $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ and $c \in \mathbb{R}$, let $f(x, y) = x^\top \mathbf{A} x + y^\top \mathbf{B} x + c$. Define

$$\nabla_z f(x, y) = \begin{bmatrix} \frac{\partial f}{\partial z_1}(x, y) & \frac{\partial f}{\partial z_2}(x, y) & \dots & \frac{\partial f}{\partial z_n}(x, y) \end{bmatrix}^\top \in \mathbb{R}^n .$$

Note: If you are unfamiliar with gradients, you may find the resources available on the course website useful. Section 4 of Zico Kolter and Chuong Do's Linear Algebra Review and Reference may be particularly helpful.

a. *[2 points]* Explicitly write out the function $f(x, y)$ in terms of the components $A_{i,j}$ and $B_{i,j}$ using appropriate summations over the indices.

$$f(x, y) = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \begin{bmatrix} A_{11} & \dots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{n1} & \dots & A_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 & \dots & y_n \end{bmatrix} \begin{bmatrix} B_{11} & \dots & B_{1n} \\ \vdots & \ddots & \vdots \\ B_{n1} & \dots & B_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + c$$

We can represent these summations over the indices as

$$f(x, y) = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j + \sum_{i=1}^{n} \sum_{j=1}^{n} y_i B_{ij} x_j + c$$

which can be rewritten as

$$f(x, y) = \sum_{i=1}^{n} x_i \sum_{j=1}^{n} A_{ij} x_j + \sum_{i=1}^{n} y_i \sum_{j=1}^{n} B_{ij} x_j + c$$

b. *[2 points]* What is $\nabla_x f(x,y)$ in terms of the summations over indices *and* vector notation?

$$\frac{\partial}{\partial x_k}(x^\top \mathbf{A}x) = 2x_k A_{kk} + \sum_{\substack{i=1 \\ i\neq k}}^{n} A_{ki}x_i + \sum_{\substack{j=1 \\ j\neq k}}^{n} A_{jk}x_j$$

$$\frac{\partial}{\partial x_k}(y^\top \mathbf{B}x) = y_k B_{kk} + y_2 B_{2k} + ... + y_n B_{nk} = \sum_{i=1}^{n} B_{ik}y_i$$

$$\nabla_x f(x,y)_k = \frac{\partial f(x,y)}{\partial x_k} = \sum_{\substack{i=1 \\ i\neq k}}^{n} A_{ik}x_i + \sum_{\substack{j=1 \\ j\neq k}}^{n} A_{kj}x_j + \sum_{i=1}^{n} B_{ik}y_i + 2A_{kk}x_k$$

Since

$$\sum_{i=1}^{n} A_{ki}x_i = x_k A_{kk} + \sum_{\substack{i=1 \\ i\neq k}}^{n} A_{ki}x_i \qquad \text{and} \qquad \sum_{j=1}^{n} A_{jk}x_j = x_k A_{kk} + \sum_{\substack{j=1 \\ j\neq k}}^{n} A_{jk}x_j$$

We get

$$\nabla_x f(x,y) = \sum_{i=1}^{n} A_{ik}x_i + \sum_{j=1}^{n} A_{kj}x_j + \sum_{i=1}^{n} B_{ik}y_i$$

In vector notation we have

$$\nabla_x f(x,y) = A^T x + Ax + B^T y = (A + A^T)x + B^T y$$

c. *[2 points]* What is $\nabla_y f(x,y)$ in terms of the summations over indices *and* vector notation?

$$\nabla_y f(x,y)_k = \frac{\partial f(x,y)}{\partial y_k} = \partial y_k \Big(y_k \sum_{j=1}^{n} B_{kj}x_j \Big) = \sum_{j=1}^{n} B_{kj}x_j$$

$$\nabla_y f(x,y) = Bx$$

A8. Show the following:

a. *[2 points]* Let $g : \mathbb{R} \to \mathbb{R}$ and $v, w \in \mathbb{R}^n$ such that $g(v_i) = w_i$. Find an expression for g such that $\text{diag}(v)^{-1} = \text{diag}(w)$.

Let matrices $\mathbf{V}$ and $\mathbf{W}$ which are diagonal and square, i.e. $\mathbf{V}, \mathbf{W} \in \mathbb{R}^{n\times n}$, be the inverse of each other such that

$$\mathbf{W} = \begin{bmatrix} w_1 & 0 \\ 0 & w_2 \end{bmatrix} = \mathbf{G}$$

$$\mathbf{V} = \begin{bmatrix} \frac{1}{w_1} & 0 \\ 0 & \frac{1}{w_2} \end{bmatrix} = \mathbf{G}^{-1}$$

$$\therefore g(v_i) = \frac{1}{w_i}$$

b. *[2 points]* Let $\mathbf{A} \in R^{n\times n}$ be orthonormal and $x \in \mathbb{R}^n$. An orthonormal matrix is a square martix whose columns and rows are orthonormal vectors, such that $\mathbf{A}\mathbf{A}^\top = \mathbf{A}^\top\mathbf{A} = \mathbf{I}$ where $\mathbf{I}$ is the identity matrix. Show that $||\mathbf{A}x||_2^2 = ||x||_2^2$.

$$||\mathbf{A}x||_2^2 = (\mathbf{A}x)^T(\mathbf{A}x) = x^T\mathbf{A}^T\mathbf{A}x = x^T\mathbf{I}x = x^Tx = ||x||_2^2$$

c. *[2 points]* Let $\mathbf{B} \in R^{n\times n}$ be invertible and symmetric. A symmetric matrix is a square matrix satisfying $\mathbf{B} = \mathbf{B}^\top$. Show that $\mathbf{B}^{-1}$ is also symmetric.

$$(\mathbf{B}^{-1})^T = (\mathbf{B}^{-1})^T\mathbf{B}^{-1}\mathbf{B} = (\mathbf{B}^{-1})^T\mathbf{B}^T(\mathbf{B}^T)^{-1} = (\mathbf{B}\mathbf{B}^{-1})^T(\mathbf{B}^T)^{-1} = (\mathbf{B}^T)^{-1}$$

Since $\mathbf{B} = \mathbf{B}^\top$

$$(\mathbf{B}^{-1})^T = \mathbf{B}^{-1}$$

d. *[2 points]* Let $\mathbf{C} \in R^{n \times n}$ be positive semi-definite (PSD). A positive semi-definite matrix is a symmetric matrix satisfying $x^\top \mathbf{C} x \geq 0$ for any vector $x \in \mathbb{R}^n$. Show that its eigenvalues are non-negative.

$$\mathbf{A}x = \lambda x$$

Let $\mathbf{C} = \lambda$:

$$x^\top \lambda x \geq 0$$

$$x^\top x \lambda \geq 0$$

Using

$$x^\top x = ||x||_2^2$$

we have

$$(||x||_2^2)\lambda \geq 0$$

Since $||x||_2^2 \geq 0$, the eigenvalues, λ, must be non-negative.

Programming

These problems are available in a .zip file, with some starter code. All coding questions in this class will have starter code. **Before attempting these problems, you will need to set up a Conda environment that you will use for every assignment in the course. Unzip the HW0-A.zip file and read the instructions in the README file to get started.**

A9. For $\nabla_x f(x, y)$ as solved for in Problem 7:

a. *[1 point]* Using native Python, implement the summation form. You can pass matrices/vectors to the `vanilla_matmul` and `vanilla_transpose` functions, but the actual operations should be done element-wise.

```
A_T_x = vanilla_matmul(vanilla_transpose(A), x)
A_x = vanilla_matmul(A, x)
B_T_y = vanilla_matmul(vanilla_transpose(B), y)

gradient_f_x = []
for i in range(len(x)):
    gradient_f_x.append(A_T_x[i] + A_x[i] + B_T_y[i])

return gradient_f_x
```

b. *[1 point]* Using NumPy, implement the vector form.

```
return np.matmul((A + A.T), x) + np.matmul(B.T, y)
```

c. *[1 point]* Report the difference in wall-clock time for parts a-b, and discuss reasons for the observed difference.

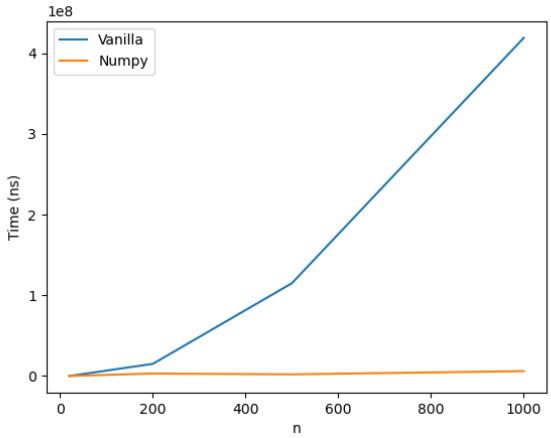


Figure 1: Wall-clock Time Between Vanilla and NumPy Implementations.

```
Time for vanilla implementation: 0.0ms
Time for numpy implementation: 0.0ms
Time for vanilla implementation: 14.9842ms
Time for numpy implementation: 2.999ms
Time for vanilla implementation: 115.0ms
Time for numpy implementation: 2.0028ms
Time for vanilla implementation: 419.1231ms
Time for numpy implementation: 5.9927ms
```

At the end of the comparison, we see a disparity of over 400 ms between the two implementations, with the NumPy implementation being significantly faster than the native Python implementation. A contributor to this difference is that NumPy arrays store vectors as contiguous blocks of memory, unlike lists in native Python which group data together that might be in distant memory locations which requires time to fetch. Also, NumPy allows for multiple vector operations to be computed in parallel, further decreasing compute time.

A10. Two random variables X and Y have equal distributions if their CDFs, F_X and F_Y, respectively, are equal, i.e. for all x, $|F_X(x) - F_Y(x)| = 0$. The central limit theorem says that the sum of k independent, zero-mean, variance $1/k$ random variables converges to a (standard) Normal distribution as k tends to infinity. We will study this phenomenon empirically (you will use the Python packages Numpy and Matplotlib). Each of the following subproblems includes a description of how the plots were generated; these have been coded for you. The code is available in the .zip file. In this problem, you will add to our implementation to explore **matplotlib** library, and how the solution depends on n and k.

a. *[2 points]* For $i = 1, \ldots, n$ let $Z_i \sim \mathcal{N}(0, 1)$. Let $F(x)$ denote the true CDF from which each Z_i is drawn (i.e., Gaussian). Define $\widehat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{Z_i \leq x\}$ and we will choose n large enough such that, for all $x \in \mathbb{R}$,

$$\sqrt{\mathbb{E}\left[\left(\widehat{F}_n(x) - F(x)\right)^2\right]} \leq 0.0025 \ .$$

Plot $\widehat{F}_n(x)$ from -3 to 3.

$$n = 40000$$

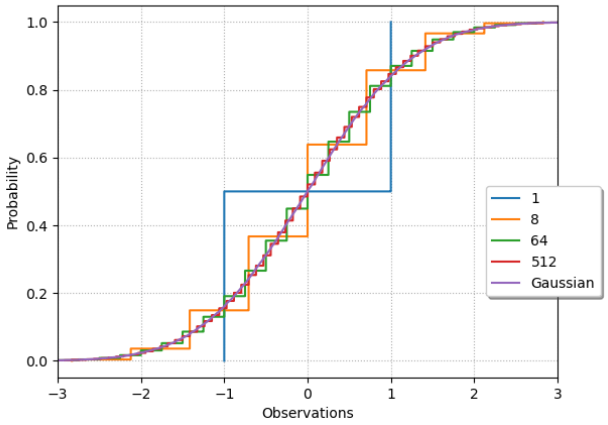


Figure 2: Empirical CDF of $\widehat{F}_n(x)$

```
plt.xlim(-3, 3)
plt.xlabel('Observations')
plt.ylabel('Probability')
plt.show()
```

b. *[2 points]* Define $Y^{(k)} = \frac{1}{\sqrt{k}} \sum_{i=1}^{k} B_i$ where each B_i is equal to -1 and 1 with equal probability and the B_i's are independent. We know that $\frac{1}{\sqrt{k}} B_i$ is zero-mean and has variance $1/k$. For each $k \in \{1, 8, 64, 512\}$ we will generate n (same as in part a) independent copies $Y^{(k)}$ and plot their empirical CDF on the same plot as part a.

The empirical CDF changes with respect to k according to the central limit theorem. As $k \to \infty$, we should see convergence toward a normal distribution. This tendency is reflected in the following plot.

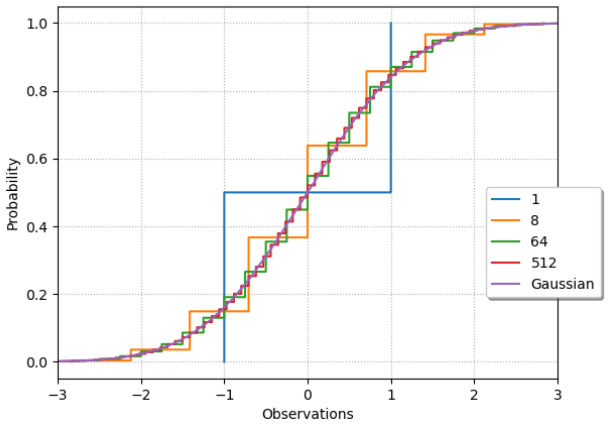


Figure 3: Empirical CDF of $\widehat{F}_n(x)$